TEXORA · LECTURE SHEET

Measurement of Linear Density of Lap, Sliver, Roving and Yarn

"Count" is the textile industry's word for how heavy a strand is for its length. This sheet sets out the direct and indirect numbering systems, the weighing methods for every strand form from lap to yarn, and the conversions and moisture corrections that keep counts honest.

TE-3103 · QUALITY EVALUATION OF FIBRE AND YARN · TOPIC 10 · LEVEL 3 TERM 1

1. The idea — mass per unit length

Strands are too irregular in cross-section for diameter to describe them (compare Topic 4). Their size is therefore stated as **linear density**: mass per unit length, or its inverse. Two families of systems grew up historically, and the tester must be fluent in both:

FAMILY	DEFINITION	SYSTEMS	BEHAVIOUR
Direct (mass per length)	Mass of a fixed standard length	**tex** = g per 1000 m; **denier** = g per 9000 m; dtex, mtex, ktex are decimal multiples	Bigger number = coarser strand
Indirect (length per mass)	Number of standard hanks in a fixed mass	**English cotton count Ne**: hanks of 840 yd per lb; **metric count Nm**: hanks of 1000 m per kg	Bigger number = finer strand

$$\text{tex} = 590.5 / \text{Ne} \;\cdot\; \text{denier} = 5315 / \text{Ne} = 9 \times \text{tex} \;\cdot\; \text{Nm} = 1.693 \times \text{Ne} \;\cdot\; \text{tex} = 1000 / \text{Nm}$$

(The constants follow from the definitions: 1 lb = 453.6 g and 840 yd = 768.1 m, so Ne 1 = 453.6 g per 768.1 m → 590.5 g per 1000 m.) The **tex system is the SI-preferred universal system**; Ne remains the everyday language of cotton spinning in Bangladesh and much of Asia — conversions must be automatic.

2. Measuring each strand form

2.1 Lap

A measured length of lap (commonly 1 m or 1 yd cut across the full width, or a full lap weighed against its known length) is weighed; result in g/m (or oz/yd). Lap uniformity is checked by weighing successive metre lengths — variation here becomes sliver variation later.

2.2 Sliver and roving

Cut measured lengths (1–5 m for sliver; longer for roving) with a measuring frame or wrap drum of known circumference, weigh on a balance, and express as g/m, ktex, or — in the traditional mill language — as the **hank**: hank number = the indirect count on the 840 yd/lb basis applied to sliver/roving (e.g. a 0.12 hank roving). Multiple lengths from different cans/positions are averaged; between-position variation is itself a quality signal.

WORKED EXAMPLE — SLIVER WEIGHT AND ROVING HANK

(a) Five metres of card sliver weigh 25.0 g → 5.0 g/m = **5.0 ktex** (5000 tex).

(b) A roving length of 15 yd weighs 30 grains. Hank = (length in hanks) / (mass in lb) = (15/840) / (30/7000) = 0.017857 / 0.004286 = **4.17 hank** — a fine roving, of the kind used for fine counts. Cross-check through tex: 30 grains = 1.944 g and 15 yd = 13.72 m, so linear density = 1.944/13.72 = 0.1417 g/m = 141.7 tex, and 590.5/141.7 = 4.17 ✓. Always cross-check an indirect result through tex before accepting it; the two routes must agree.

2.3 Yarn — the wrap reel and the lea

1. A **wrap reel** of standard circumference (1.5 yd for cotton lea work — giving 120 yd in 80 turns… precisely: the standard lea is 120 yd; reel circumference and turn count are chosen to deliver it) winds a **lea (skein)** under standard tension.
2. The lea is weighed on a balance (or a count balance / quadrant balance that reads count directly from the lea's weight).
3. Count is computed from length and mass; several leas per package and several packages per lot are averaged, and the **CV% of count** reported alongside the mean.

$$\text{Ne} = (\text{length in yards} / 840) / (\text{mass in lb}) \quad \cdot \quad \text{tex} = (\text{mass in g} / \text{length in m}) \times 1000$$

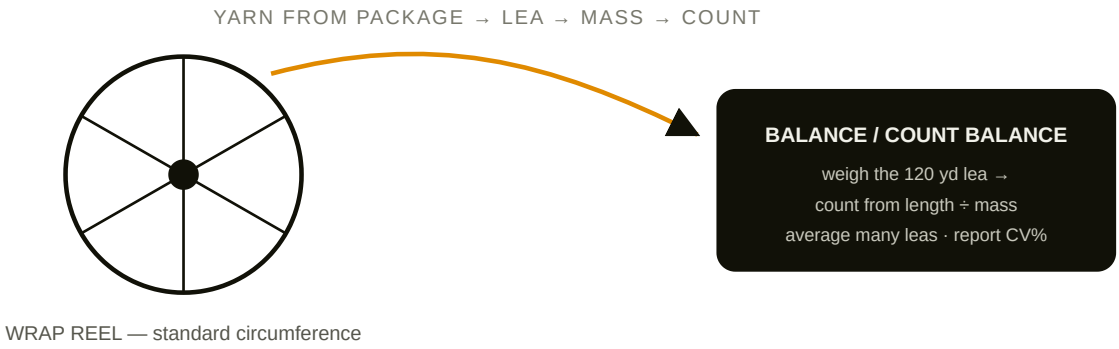


Figure 1 — Wrap reel and balance. The reel guarantees the length; the balance supplies the mass; the count is their ratio — nothing else enters.

WORKED EXAMPLE — YARN COUNT FROM A LEA

A lea of 120 yd weighs 35.0 grains. Mass = 35.0/7000 = 0.00500 lb. Length = 120/840 = 0.14286 hank.

Ne = 0.14286 / 0.00500 = **28.6**. In tex: 35 grains = 2.268 g; 120 yd = 109.7 m; tex = 2.268/109.7 × 1000 = **20.7 tex**. Cross-check: 590.5/28.6 = 20.6 tex ✓ (rounding). A mill calling this yarn "30s" is spinning 1.4 counts coarse — about 5 % extra material in every fabric — which is why count is a tested specification, not a label.

3. Moisture correction of count

Count is a mass ratio, so the regain of the yarn at weighing enters it directly. Counts are compared at the **standard regain** of the fibre (cotton 8.5 %, Topic 2). If yarn tested at regain R_1 gives count C_1 (indirect), the count at standard regain R_s is:

$$C_s = C_1 \times (100 + R_1) / (100 + R_s)$$

WORKED EXAMPLE — CORRECTING A COUNT TO STANDARD REGAIN

Cotton yarn tests at Ne 28.6 with the yarn at 6.5 % regain (a dry day, unconditioned packages). Corrected to 8.5 %:

$C_s = 28.6 \times (106.5)/(108.5) = 28.6 \times 0.9816 =$ **28.1**.

Direction check: at standard regain the yarn will hold more water, i.e. more mass per length, so the indirect count (length per mass) must *fall* ✓. Testing conditioned yarn in the standard atmosphere makes this correction small — which is precisely why standards require it.

4. Count variation

Mean count alone hides the commercial risk. Report:

- **Within-package variation** — lea-to-lea along one cone (process drift, piecing errors).
- **Between-package variation** — cone to cone (frame-to-frame and doff differences).
- **CV% of count** — good ring-spun cotton yarns show lea-count CV around 1.5–3 %; a rising CV with a correct mean points at one bad position or frame, not at the whole mill.
- Short-length variation (Topic 13) is a different, finer-scale phenomenon measured by evenness testers; the lea averages 120 yd and smooths it away.

Glossary

Linear density — mass per unit length. **tex** — grams per 1000 m (direct). **Denier** — grams per 9000 m. **Ne** — English cotton count: 840 yd hanks per lb (indirect). **Nm** — metric count: 1000 m hanks per kg. **Lea / skein** — the standard 120 yd length of yarn wound for weighing. **Wrap reel** — the standard-circumference reel that winds the lea. **Hank (sliver/roving)** — the indirect count applied to intermediate strands.

Key points

• Direct systems (tex, denier): bigger = coarser. Indirect (Ne, Nm): bigger = finer. Convert through 590.5/Ne = tex.
• Every count is a weighing: fixed length on the reel or frame, mass on the balance.
• Correct counts to standard regain (cotton 8.5 %) before judging them.
• Report mean count with CV% and sample structure (leas per package, packages per lot).

Practice

1. Convert Ne 40 to tex, denier and Nm.
2. A 120 yd lea weighs 2.00 g. Find the count in Ne and in tex, showing both routes.
3. 4 m of roving weighs 0.62 g. Find its linear density in tex and its hank.
4. Yarn at 10.0 % regain tests Ne 30.0. Correct the count to cotton's standard regain.

Answer key

1. tex = 590.5/40 = **14.8 tex**; denier = 9 × 14.8 = **132.8 den** (5315/40 = 132.9 by the other constant — rounding); Nm = 1.693 × 40 = **67.7**.
2. tex route: 120 yd = 109.7 m; tex = 2.00/109.7 × 1000 = **18.2 tex**. Ne = 590.5/18.2 = **32.4**. Direct Ne check: mass = 2.00 g = 30.86 grains = 0.004409 lb; length = 0.14286 hank; Ne = 0.14286/0.004409 = 32.4 ✓.

3. tex = 0.62/4 × 1000 = **155 tex**. Hank (Ne basis) = 590.5/155 = **3.81 hank**.
4. C_s = 30.0 × (110.0)/(108.5) = **30.4**. The yarn was wetter than standard, hence heavier per length, hence its as-tested indirect count was depressed; at standard regain it is finer than it appeared.
